# Practice Paper

**MATHEMATICS** **Syllabus 9709**

Paper 3 Pure Mathematics 3 **Written in the 2026 style**

**1 hour 50 minutes**

You must answer on separate answer paper.

You will need: Answer paper
List of formulae (MF19)

**INSTRUCTIONS**

- Answer **all** questions.
- Write your answers on the answer paper. Write the question number next to each answer.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

**INFORMATION**

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

*This practice paper was written to show the features of the 2026 question papers: questions that join two or more topics, results in terms of general constants, hints that are less direct, and short contexts. It is not an official Cambridge International document.*

This document has 6 pages.

**[Turn over**

**1** Find the set of values of $x$ satisfying the inequality $|3x - 2a| > 2|x + a|$, where $a$ is a positive constant. [4]

**2** It is given that

$$\int_0^a (3e^{2x} - 2e^x)\,dx = 8,$$

where $a$ is a positive constant.

**(a)** Show that $3e^{2a} - 4e^a - 15 = 0$. [3]

**(b)** Hence find the exact value of $a$. [2]

**3** **(a)** Show that

$$p^3 + q^3 \equiv (p + q)^3 - 3pq(p + q).$$ [1]

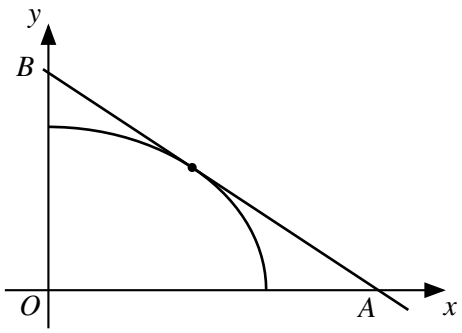
**(b)** Hence, or otherwise, solve the equation

$$\sin^6 x + \cos^6 x = \frac{7}{16} \qquad \text{for } 0 < x < \pi.$$

Give your answers in exact form. [5]

**4** A curve has parametric equations

$$x = 4\cos\theta, \qquad y = 3\sin\theta, \qquad \text{for } 0 < \theta < \tfrac{1}{2}\pi.$$




**(a)** Show that the equation of the tangent to the curve at the point with parameter $\theta$ is

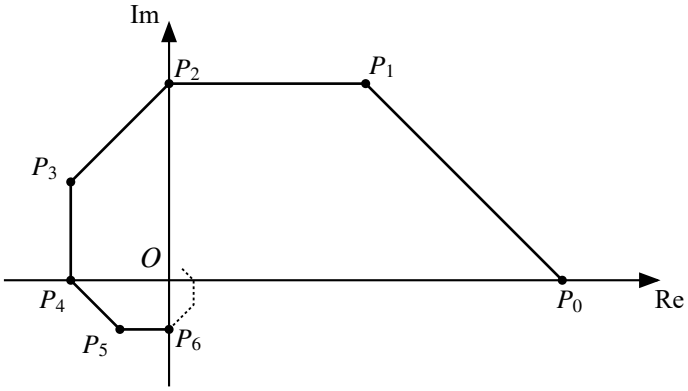
$$3x\cos\theta + 4y\sin\theta = 12.$$ [4]

The tangent meets the $x$-axis at $A$ and the $y$-axis at $B$, as shown in the diagram. The point $O$ is the origin.

**(b)** Given that the area of triangle $AOB$ is 16 square units, find the possible values of $\theta$. [4]

**5** The complex number $w$ is given by $w = \frac{1}{2}(1 + \mathrm{i})$.

**(a)** Express $w$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leqslant \pi$. Give the exact values of $r$ and $\theta$. [2]




On an Argand diagram with origin $O$, the points $P_0, P_1, P_2, P_3, \ldots$ represent the complex numbers $1, w, w^2, w^3, \ldots$ respectively. The diagram shows the points $P_0$ to $P_6$ joined in order by straight lines.

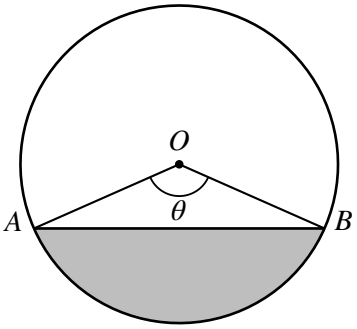
**(b)** Show that the lengths $P_0P_1, P_1P_2, P_2P_3, \ldots$ form a geometric progression.

Hence find the exact total length of the path $P_0P_1P_2P_3\ldots$ as the number of points tends to infinity. Give your answer in the form $a + b\sqrt{2}$, where $a$ and $b$ are integers. [4]

**(c)** Find the smallest value of $n$ for which $P_n$ lies inside the circle $|z| = 0.01$. [2]

**[Turn over**

**6**




The diagram shows the cross-section of a horizontal cylindrical tank of radius 1 m. The centre of the cross-section is $O$. The tank contains some water, shown shaded. The surface of the water meets the circle at $A$ and $B$, and angle $AOB = \theta$ radians, where $0 < \theta < \pi$.

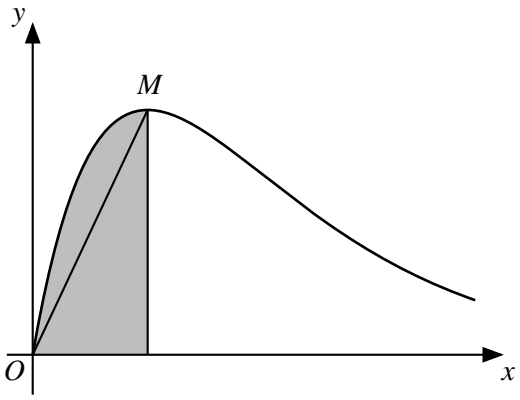
The tank is one quarter full of water.

**(a)** Show that $\theta = \frac{1}{2}\pi + \sin\theta$. [2]

**(b)** Show by calculation that $\theta$ lies between 2.2 and 2.4. [2]

**(c)** Use an iterative formula based on the equation in **6(a)** to find $\theta$ correct to 2 decimal places. Give the result of each iteration to 4 decimal places. Hence find the depth of the water in the tank. [4]

**7**




The diagram shows the curve with equation $y = x\mathrm{e}^{-\frac{1}{2}x}$, for $x \geqslant 0$, and its maximum point $M$. The region $R$, shaded in the diagram, is bounded by the curve, the $x$-axis and the line through $M$ parallel to the $y$-axis.

**(a)** Find the exact coordinates of $M$. [3]

**(b)** The line $OM$ divides $R$ into two regions.

Show that the area of the region above $OM$ and the area of the region below $OM$ are in the ratio $(2\mathrm{e} - 5) : 1$. [6]

**8** In a model for the number of fish, $N$, in a lake, $t$ years after the fish are first counted, $N$ and $t$ satisfy the differential equation

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \frac{(N - 200)(800 - N)}{5000}.$$

$N$ is treated as a continuous variable. When the fish are first counted, there are 500 fish in the lake.

**(a)** Express $\dfrac{1}{(N - 200)(800 - N)}$ in partial fractions. [2]

**(b)** Solve the differential equation, obtaining an expression for $N$ in terms of $t$. [5]

**(c)** State what happens to the number of fish in the lake in the long term, according to this model.

In a second lake, the number of fish satisfies the same differential equation, but there are 150 fish when they are first counted. Without solving the differential equation, explain what the model predicts for the number of fish in the second lake. [2]

**9** With respect to the origin $O$, the points $A$ and $B$ have position vectors $5\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$ and $2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ respectively. The line $m$ passes through $A$ and $B$.

**(a)** Find a vector equation for $m$. [2]

The line $l$ has equation

$$\mathbf{r} = \mathbf{i} + 4\mathbf{k} + \lambda(2\mathbf{i} - \mathbf{j} + 2\mathbf{k}).$$

**(b)** Show that $l$ and $m$ intersect, and find the position vector of the point of intersection, $C$. [3]

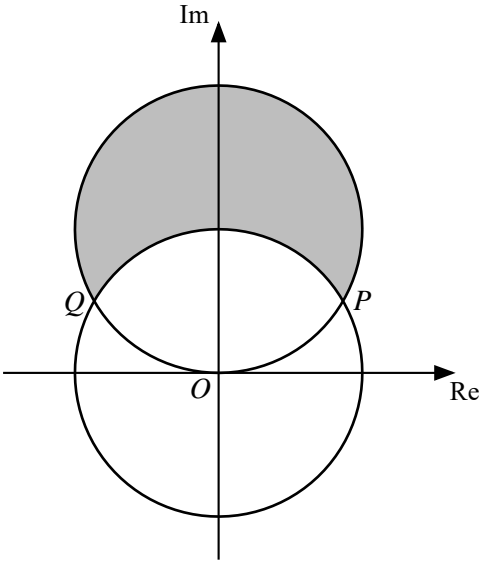
**(c)** The point $D$ lies on $l$ and is such that $AD$ is perpendicular to $l$.

Find the exact area of triangle $ACD$. [4]

**Question 10 is on the back page.**

**[Turn over**

**10**




The diagram shows a design for a logo, drawn on an Argand diagram. The logo is the shaded region $R$, whose points represent the complex numbers $z$ satisfying both $|z - 2i| \leqslant 2$ and $|z| \geqslant 2$.

The circles $|z - 2i| = 2$ and $|z| = 2$ meet at the points $P$ and $Q$.

**(a)** Find the complex numbers represented by $P$ and $Q$. Give your answers in the form $x + iy$, where $x$ and $y$ are real and exact. [3]

**(b)** Find the greatest value of $\arg z$ for points in $R$. Give your answer in exact form. [2]

**(c)** Find the exact area of $R$. [4]